

Personal AI Agent Design Document: Study Coach & Tracker

1. Job to Be Done:

Personal Study Coach and Academic Progress Tracker. Ingests coursework notes and schedule to answer complex questions, generate quizzes, and track milestones.

2. User & Usage Frequency:

Undergraduate Software Engineering student; daily active usage (15-30 mins/day).

3. Tools, Data Needed & Access Plan:

- Data Sources: Local Markdown/PDF lecture notes and university course outlines.
- Tools: Filesystem Reader Tool & SQLite Database Connector.
- Access Plan: Local read/write permissions scoped strictly to project directories.

4. Draft Agent Instructions:

'You are an expert AI Study Coach grounded exclusively in the user's local academic notes. Maintain an encouraging, direct, and academically rigorous tone.'

5. Pre-Build Eval Cases:

1. Accurate Retrieval from Notes: Verify responses use exact local definitions.
2. Missing Data Handling: Acknowledge missing files without hallucination.
3. Quiz Generation: Produce tailored practice questions matching syllabus difficulty.
4. Schedule Conflict Resolution: Prioritize tasks based on weight and deadlines.
5. Guardrail Test: Refuse to solve active graded exam questions directly.

6. Risks, Guardrails & Boundaries:

- Must Confirm: Any automated rescheduling or milestone log modifications.
- Must Never Do: Output direct solutions to active graded exams or modify system files.

7. Platform Choice & Justification:

Claude Project with Filesystem Connectors. Chosen over generic GPTs for secure, direct local file access while preserving data privacy over personal university materials.